Lecture 12: Special Cases of the Dependent Variable

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

Four cases of the same equation

01

THE COMMON FRAME

What stays fixed and what changes when the type of Y changes.

02

SKEWED Y: LOG MODELS

ln(Y) on the left; coefficients read as percentage changes.

03

BINARY Y: PROBABILITY MODELS

Y = 0/1; the linear probability model and logistic regression.

04

Y OVER TIME

Observations ordered in time; the independence assumption breaks.

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The invariant right-hand side

The right-hand side $\beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k$ stays the same. What changes is how Y enters the left-hand side, and therefore how each β is read.

The interpretation "holding the other X constant" carries over to every case in this block.

THE COMMON FRAME

Map of the cases

Type of Y	Left-hand side	β is read as	Estimated by
Continuous, symmetric	Y	Change in Y, in its units	OLS
Continuous, right-skewed	ln(Y)	Percentage change in Y	OLS
Binary (0/1)	P(Y = 1)	Change in probability	OLS (LPM)
Binary (0/1)	ln(odds)	Odds ratio, via e^β	Logistic regression
Observed over time	Y_t	As above; errors not independent	OLS with time terms
All cases	Chosen by the type of Y	Read by the left-hand side	Same right-hand side

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SKEWED Y: LOG MODELS

Log-linear model

$\ln(Y) = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k + \varepsilon$

MECHANISM

Logarithms turn ratios into differences

$\ln(1.10 \times Y) - \ln(Y) = \ln(1.10) \approx 0.10$. A constant difference in ln(Y) is a constant percentage change in Y. Taking logs also pulls in the long right tail.

CONSEQUENCE

Coefficients become percentages

A one-unit increase in X changes Y by about 100·β percent. For a dummy, the exact effect is (e^β − 1) × 100 percent.

- Small β (below about 0.1): 100·β is a good approximation
- Large β or dummies: use (e^β − 1)

Why might a 1-point higher score add a fixed percentage rather than a fixed number of dollars?

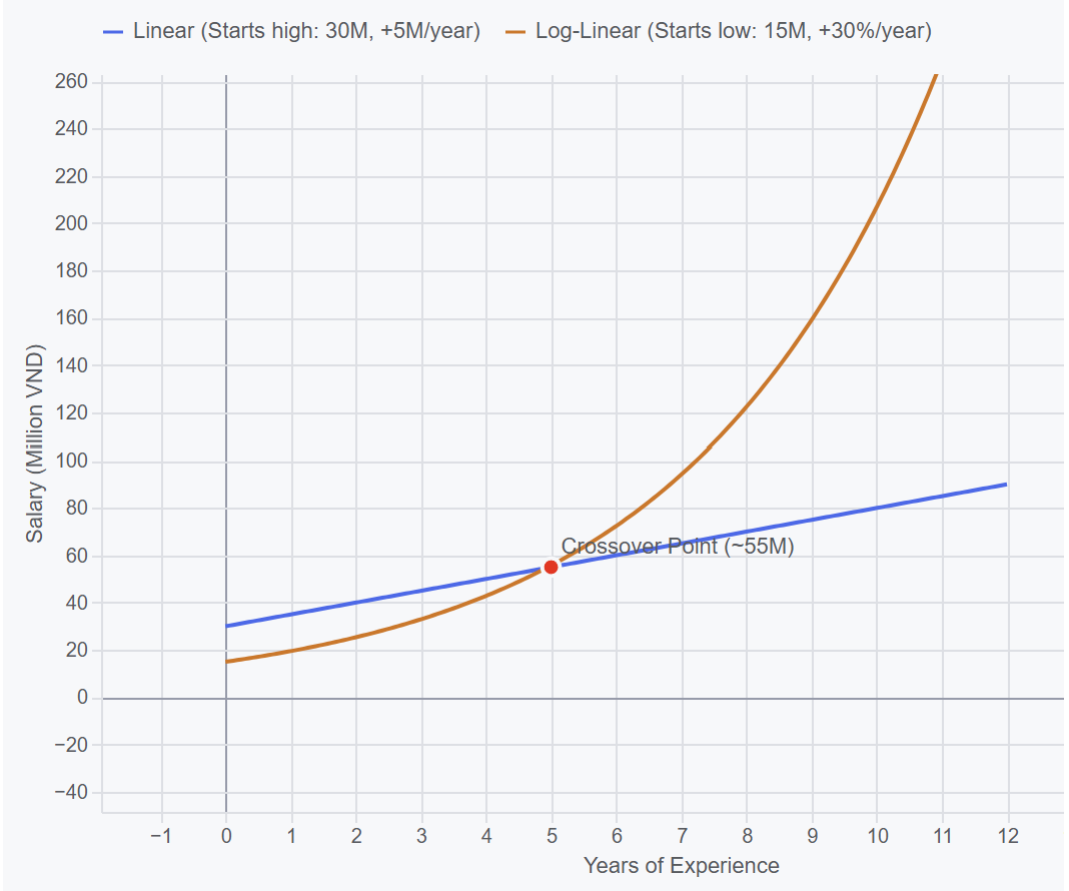
Why use log-linear model

$$\ln(Y) = \beta_o + \beta_1 X_1 + \ldots + \beta_k X_k + \varepsilon$$

Log-transformed dependent variable models

A log-linear model is often used when the relationship between the variables is multiplicative rather than additive

- Growth or Changes are Percentage-Based (Exponential)
- Highly Skewed Data (Non-Normality)
- Heteroskedasticity – Không đồng nhất phương sai (Unstable Variance)
- Modeling "Elasticity" in Economics (Log-Log Variant)



Four functional forms

Model	Equation	Reading of β_1
Linear	$Y = \beta_0 + \beta_1 X$	X + 1 unit → Y changes by β_1 units
Log-linear	$\ln(Y) = \beta_0 + \beta_1 X$	X + 1 unit → Y changes by about $100 \cdot \beta_1$ %
Linear-log	$Y = \beta_0 + \beta_1 \ln(X)$	X + 1 % → Y changes by about $\beta_1/100$ units
Log-log	$\ln(Y) = \beta_0 + \beta_1 \ln(X)$	X + 1 % → Y changes by about β_1 % (elasticity)
Rule	A log on one side	That side is read in percent

Interpreting Log-linear model

The Quick Approximation Method

$$\%\Delta Y \approx \beta_1 \times 100\%$$

The Exact Method

$$\%\Delta Y = \left(e^{\beta_1} - 1\right) \times 100\%$$

Model Type	Dependent (Y)	Independent (X)	Change in X	Expected Change in Y
Level–Level	Y	X	1 unit increase	β_1 units change
Log–Level	$\ln(Y)$	X	1 unit increase	$\left(e^{\beta_1} - 1\right) \times 100\%$ change
Level–Log	Y	$\ln(X)$	1% increase	$(\beta_1/100)$ units change
Log–Log	$\ln(Y)$	$\ln(X)$	1% increase	β_1 %change

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Linear probability model (LPM)

OLS with a 0/1 dependent variable: $P(Y=1) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots$

MECHANISM

The mean of a 0/1 variable is a probability

When Y takes only the values 0 and 1, $E[Y \mid X] = P(Y = 1 \mid X)$. OLS on Y therefore estimates a probability as a straight line in X.

CONSEQUENCE

β is a change in probability

A one-unit increase in X changes $P(Y = 1)$ by β, measured in percentage points.

- Easy to estimate and read in Excel
- Fitted probabilities can fall below 0 or above 1
- The effect of X is forced to be the same at every level

Logistic regression

$\ln(p/(1-p)) = \beta_0 + \beta_1 X_1 + \ldots + \beta_k X_k$, with $p = P(Y = 1 \mid X)$

MECHANISM

The model is linear in the log-odds

The odds p/(1 − p) can take any positive value, and their logarithm any real value. The right-hand side is fitted to ln(odds), not to p itself.

CONSEQUENCE

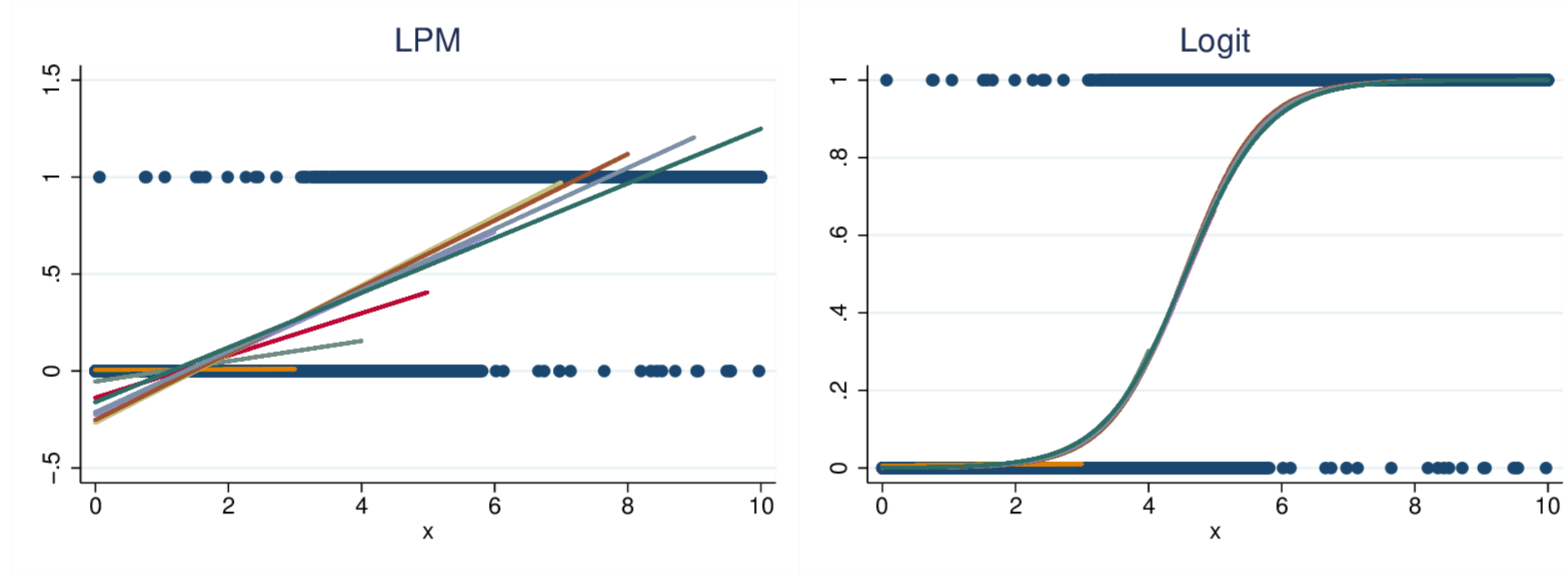
β is read through the odds ratio

A one-unit increase in X multiplies the odds of Y = 1 by e^β.

- Fitted probabilities always lie between 0 and 1
- Estimated by maximum likelihood, not by the Excel ToolPak
- Output reports z-statistics and a pseudo-R^2 instead of t and R^2

- $\beta > 0$: higher X **increase** log-odds (the probability of the event happening $Y = 1$).
- $\beta < 0$: higher X **decrease** log-odds (the probability of the event happening $Y = 1$).
- $\beta = 0$: X has **no effect** on the outcome.
- Odds Ratio (OR) $= e^{\beta}$: For every 1 unit increase in X, the odds of Y happening **increase by (OR – 1)**

Linear Probability Model vs Logistic regression



Test for Goodness of Fit

Does the proportion follow the expected value?

Reality vs Theory

O is the real data you actually collected.

E is the theoretical data you expect to see if there is absolutely NO connection between the two variables

Cells	Observed Frequency (O)	Probability under H0 (p0)	Expected Frequency (E)	Component of χ2
Group 1	n_1	p_{10}	$E_1 = n \cdot p_{10}$	$\frac{(n_1 - E_1)^2}{E_1}$
Group 2	n_2	p_{20}	$E_2 = n \cdot p_{20}$	$\frac{(n_2 - E_2)^2}{E_2}$
…	…	…	…	…
Group k	n_k	p_{k0}	$E_k = n \cdot p_{k0}$	$\frac{(n_k - E_k)^2}{E_k}$
Total	n	1	n	$\chi^2 = \sum \frac{(O - E)^2}{E}$

The Pearson Chi-squared Test, Step by Step

STEP 1

Hypotheses

$H_0: p_1 = p_{10}, p_2 = p_{20}, \dots, p_k = p_{k0}$

H_1: At least one $p_i \neq p_{i0}$ with $i = 1, \dots, k$

STEP 2

Test statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(n_i - np_{i0})^2}{np_{i0}}$$

STEP 3

Rejection region

Calculated using α and $d.f. = k - 1$

$$\chi^2_{\alpha,k-1}$$

STEP 4

Compute

$\chi^2 = \sum_{i=1}^{k} \frac{(n_i - np_{i0})^2}{np_{i0}} = \sum_{\text{cells}} \frac{(O-E)^2}{E}$

STEP 5

Conclude

Compare χ^2 with $\chi^2_{\alpha,k-1}$ to decide whether to reject H_0.

Reject if $\chi^2 \geq \chi^2_{\alpha,k-1}$

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Y OVER TIME

Time-series regression

$Y_t = \beta_0 + \beta_1 X_{1t} + \ldots + \varepsilon_t$, with $t = 1, 2, \ldots, T$

MECHANISM

Neighbouring observations share shocks

When the same unit is observed period after period, a shock in period t usually persists into t + 1. The errors ε_t and ε_{t+1} are then correlated.

CONSEQUENCE

Standard errors are no longer reliable

The usual SE assumes independent errors; with correlated errors it is typically too small, so t-statistics are too large and p-values too small.

- Add the lagged value Y_{t-1} as a regressor
- Add a time trend and seasonal dummies

Independence is the assumption behind every standard error in this course; time series is where it fails first.